

This Patient Group Direction (PGD) is intended only for registered healthcare professionals who have been named and authorised by their organisation to practice under it.

Patient Group Direction
for the administration of Hydroxocobalamin for the treatment of Vitamin B12 Deficiency

Summary of NICE CKS Guidelines for Low Vitamin B12

Overview

- Vitamin B12 deficiency can cause **megaloblastic anaemia**, **neurological symptoms**, and **non-specific symptoms** (e.g. fatigue).
- Causes include **pernicious anaemia**, **malabsorption**, **dietary deficiency**, and **medications** (e.g. metformin, PPIs).

Symptoms

General:

- Fatigue
- Weakness
- Dyspnoea
- Pale or jaundiced skin

Neurological:

- Paraesthesia (numbness/tingling)
- Peripheral neuropathy
- Cognitive changes
- Ataxia

Haematological:

- Macrocytic anaemia
- Glossitis
- Low white blood cells/platelets (in severe cases)

Diagnosis

- **Check serum B12 levels** if:
 - Symptoms suggestive of deficiency.
 - Macrocytic anaemia on FBC.
 - Risk factors present (e.g. vegan diet, elderly, GI surgery).
- Also assess:
 - **Folate and ferritin** (other deficiencies may coexist).
 - **Intrinsic factor antibodies** (for pernicious anaemia).
 - **MMA and homocysteine** (specialist settings) if B12 borderline.

Note: Normal B12 levels can still occur in symptomatic patients — clinical judgement is key.

Causes

- **Pernicious anaemia** (autoimmune destruction of intrinsic factor).
 - **Dietary deficiency** (especially in vegans).
 - **Malabsorption** (e.g. post-gastrectomy, Crohn's disease).
 - **Medications** (e.g. metformin, PPIs, H2 blockers).
 - **Alcohol misuse, elderly, GI disorders.**
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Management

If symptomatic or confirmed B12 deficiency:

1. Hydroxocobalamin injections (first-line in the UK):

- **1 mg IM every other day for 2 weeks** (initial correction phase).
- Then:
 - **Every 3 months for life** if due to **pernicious anaemia or irreversible cause**.
 - **Stop after correction** if reversible (e.g. dietary), and re-evaluate oral intake.

2. Oral B12 (cyanocobalamin):

- For **dietary deficiency only** where absorption is intact.
 - Dose: 50–150 mcg daily (higher doses may be used off-label).
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Monitoring and Follow-Up

- Check symptom improvement after 4–6 weeks.
- Repeat FBC after initial treatment if anaemia was present.
- Monitor neurological symptoms closely — may take **months to resolve**.
- Lifelong monitoring if cause is permanent.

When to Refer

- Diagnostic uncertainty.
- Neurological symptoms not improving.
- Borderline B12 with normal haematology but persisting symptoms.
- Suspected malabsorption or coexisting conditions (e.g. coeliac disease).

Interpreting blood test results (guidance for pharmacists)

- Investigation of suspected deficiency should include a full blood count, blood film, serum folate, and either total B12 (serum cobalamin) or active B12 (serum holotranscobalamin). Active B12 is the preferred marker where available, as it avoids false low results in women taking oral contraceptives and in pregnancy.
- Full blood count: an MCV greater than 100 femtolitres indicates macrocytosis, but MCV may be normal with coexisting iron deficiency or rapidly developing anaemia. A normal MCV does not exclude the need for cobalamin testing; neurological impairment occurs with a normal MCV in around 25% of cases. White cells and platelets may be reduced if anaemia is severe, and the reticulocyte count may be low relative to the degree of anaemia.
- Blood film: oval macrocytes and hypersegmented neutrophils (more than 5% of neutrophils with five or more lobes, or any neutrophil with six or more lobes) suggest B12 or folate deficiency, though these features are neither sensitive nor specific in early deficiency.
- B12 thresholds (NICE): confirmed deficiency — total B12 below 180 nanograms/L (133 picomol/L) or active B12 below 25 picomol/L. Possible deficiency — total B12 180 to 350 nanograms/L or active B12 25 to 70 picomol/L. Deficiency unlikely — total B12 above 350 nanograms/L or active B12 above 70 picomol/L.
- Interpret results in clinical context: cobalamin levels correlate poorly with symptoms; clinically significant deficiency can exist with levels in the normal range, particularly in the elderly. People already taking B12 may have raised levels without deficiency being treated. Levels fall in pregnancy and are less reliable there.
- Consider serum methylmalonic acid (MMA) where symptoms or signs of deficiency accompany an indeterminate total or active B12 result.
- Be aware that people of Black ethnicity may have a higher reference range for serum vitamin B12 than people of White or Asian ethnicity.
- Folate: serum folate below 7 nanomol/L (3 micrograms/L) suggests deficiency, with an indeterminate zone of 7 to 10 nanomol/L — suggestive, not diagnostic. If clinical suspicion is strong with normal serum levels, red cell folate can be measured once cobalamin deficiency has been excluded; below 340 nanomol/L (150 micrograms/L) is consistent with folate deficiency in the absence of B12 deficiency.

Patient Group Direction

for the supply/administration of

Hydroxocobalamin 1mg

for the treatment of

Vitamin B12 deficiency

Healthcare professionals covered and training

	Healthcare professionals allowed to work under this PGD and requirements
Qualifications and Registration	● Pharmacist registered and practicing with the GPhC ● Pharmacy technician registered and practicing with the GPhC
Training and assessment	● Pharmacists and Pharmacy Technicians must have completed training relevant to this condition to use this PGD. This training will be documented and overseen by the Get Real Health team. ● All users must be familiar with the SPC for the products as well as BNF advice and any applicable national guidelines ● All users must previously have used PGD's to supply medication ● Must work in compliance with the SOPs of their own employer ● All users must have appropriate indemnity in place to cover this service
Further training and responsibilities	● All users must be up to date with CPD requirements and appraisal ● All users must be up to date with MHRA and safety alerts with regards to medical products ● All users must understand the concepts of capacity and consent and be aware of the Mental Capacity Act 2005.

PGD Indication	Hydroxocobalamin is indicated for the treatment of vitamin B12 deficiency, including pernicious anaemia, malabsorption, or dietary deficiency. It supports neurological function, red blood cell formation, and DNA synthesis.
Inclusion criteria	- Confirmed diagnosis of vitamin B12 deficiency (e.g., blood tests, clinical symptoms). - Adults aged 18 years or older. - No contraindications to hydroxocobalamin. - Informed consent obtained. - Suitable for intramuscular injections in a non-acute setting.
Exclusion criteria	- Known hypersensitivity to hydroxocobalamin or cobalamin derivatives. - Active infection at proposed injection site. - Known hereditary problems of cobalamin metabolism. - Severe thrombocytopenia or bleeding disorders unless managed appropriately. - Individuals with Leber's disease (risk of optic atrophy). Pregnancy or breastfeeding — refer to the GP or midwife; B12 test interpretation is unreliable in pregnancy and management belongs with the antenatal team.
Cautions	- Monitor potassium levels during initiation in severe deficiency (risk of hypokalaemia). - Inform patients that urine may turn red post-injection. - Rare anaphylactic reactions may occur — observe after administration. - Use with caution in individuals with history of allergies or asthma.
Actions if patient is excluded or declines treatment	Advise on alternative treatment options and how these can be accessed. Document any advice given and the decision reached. Inform or refer to the GP as appropriate

Details of the medicine

Name, form and strength of medicine	Hydroxocobalamin 1mg
Legal category	POM
Storage	Store below 25°C. Protect from light.
Route/method of administration	Intramuscular injection.
Dose and frequency	Initial correction: 1 mg intramuscularly three times a week for 2 weeks. Maintenance: 1 mg every 2 to 3 months depending on underlying cause and response.
Quantity to be administered and/or supplied	1 vial of 1mg

Maximum or minimum treatment period	Initial correction followed by long-term maintenance as clinically indicated.
Adverse effects	Injection site reactions, dizziness, nausea, headache, rash, and rarely anaphylaxis. Urine may appear reddish post-injection. Refer to SmPC.

Yellow card reporting	Report suspected adverse effects via https://yellowcard.mhra.gov.uk and inform the GP if appropriate.
Records to be kept	Record the following information: ● that valid informed consent was given ● name of individual, address, date of birth and GP ● name of healthcare practitioner ● name and brand of medication ● date of supply dose, form and route quantity supplied ● advice given, including advice given if excluded or declines treatment ● details of any adverse drug reactions and actions taken supplied via PGD Records should be signed and dated. All records should be clear, legible and contemporaneous. Electronic records can be made. A record of all individuals receiving treatment under this PGD should also be kept for audit purposes as follows: ● Adults aged 18 years and over - keep records for 8 years ● Children aged under 18 years -keep records until the 25th birthday (if the patient was 17 years when treatment was finished, keep records until the 26th birthday)

Patient information

Written information to be given to patient or carer	Supply the patient information leaflet (PIL) provided with the medication.
Follow-up advice to be given to patient or carer	To seek medical advice if symptoms worsen rapidly or significantly, do not improve in 3–4 weeks, or they become systemically very unwell.

Key references

- NICE Medicines Practice Guideline 2 (MPG2): Patient Group Directions. Published March 2017 <https://www.nice.org.uk/guidance/mpg2>
- NICE MPG2 Patient group directions: competency framework for health professionals using patient group directions. Updated March 2017 <https://www.nice.org.uk/guidance/mpg2/resources>
- NICE CKS Guidance
- British National Formulary (BNF) Key information on the selection, prescribing, dispensing and administration of medicines [BNF \(British National Formulary\) | NICE](#)
- Summary of Product Characteristics

Agreement to practise by the Registered Healthcare Professional

By signing this PGD you are confirming that you agree to its contents and that you will work within it. PGDs do not remove inherent professional obligations or accountability. It is the responsibility of each healthcare professional to practise only within the bounds of their own competence and professional code of conduct.

Name and address of pharmacy premises / clinic premises to which this PGD relates			
Name of healthcare professional	Job title	Registration number	Signature
Valid from date		Expiry date	

PGD adoption and authorisation by the employer/clinical lead

The employer has ensured that the person using this PGD, has the knowledge and skills to safely provide the service which will encompass the supply or administration of a medication or medicine.

This section must be signed by the superintendent or clinical lead responsible for this service.

Name of superintendent / clinical lead	Job title & name of organisation	Signature	Date
Chris Pilkington	Head Pharmacist, Get Real Health Limited (GPhC: 2046322)		18/06/2026
Name of professional group signatory	Job title & name of organisation	Signature	Date



Chris Pilkington

Head Pharmacist, Get Real
Health Limited (GPhC:
2046322)

A handwritten signature in dark ink, appearing to read 'C Pilkington', written in a cursive style.

18/06/20
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Signed on behalf of Get Real Health

Name	Qualification	Signature	Date
Nitin Shori	Medical Doctor GMC: 6047293		25/8/25
Chris Pilkington	Pharmacist GPhC: 2046322		25/8/25

Change history

Version number	Change details	Date
001	Development & issue of new PGD	25 th June 2025
002	Added pharmacist guidance on interpreting blood tests (CKS/NICE thresholds; active B12 preferred where available), recommended investigation set, and pregnancy/breastfeeding exclusion. PPH clinical review (J. Wilkins) with C. Pilkington.	29th July 2026

